

NarraFlow

Turn your voice into an illustrated story.

AssemblyAI Voice Agent Hackathon 2026

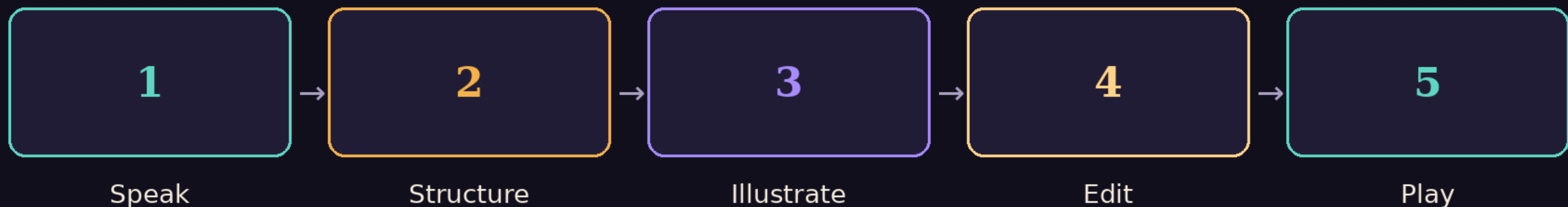
THE PROBLEM

Storytelling is natural.
Visual creation is not.
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Visual creation is not.

- People tell stories effortlessly — a parent at bedtime, a teacher, a child.
- Turning that into pictures usually means writing prompts, or drawing.
- That gap keeps great stories stuck in people's heads.

THE SOLUTION

Speak → Structure → Illustrate → Edit → Play



- Press one microphone button and just talk — no prompts to write.
- AssemblyAI's Voice Agent understands narration vs. instructions.
- Every scene is illustrated live; characters keep the same look throughout.

The whole conversational layer

- Voice Agent API — real-time speech recognition, turn-taking, spoken replies
- Barge-in — interrupt the agent mid-sentence; it stops and listens
- Structured tool calling — 13 tools turn speech into real story actions
- Server-issued, single-use tokens — the permanent API key never reaches the browser

VOICE → ASSEMBLYAI → TOOL CALL → STORY ENGINE → ILLUSTRATION → SPOKEN REPLY

Every arrow in that chain runs on live AssemblyAI infrastructure.

WHAT MAKES NARRAFLOW DIFFERENT

Not another voice chatbot

Character memory

Each character's look is stored once and reused in every illustration prompt.

Selective regeneration

Editing a character only repaints scenes that contain them — not the whole story.

Voice editing

“Make the tree blue.” “Change her dress to green.”
Changes happen by talking.

Illustrated playback

The finished story plays back with captions, narration and an exportable video.

USE CASES

Who it's for

Education

Language learning

Accessibility

Family storytelling

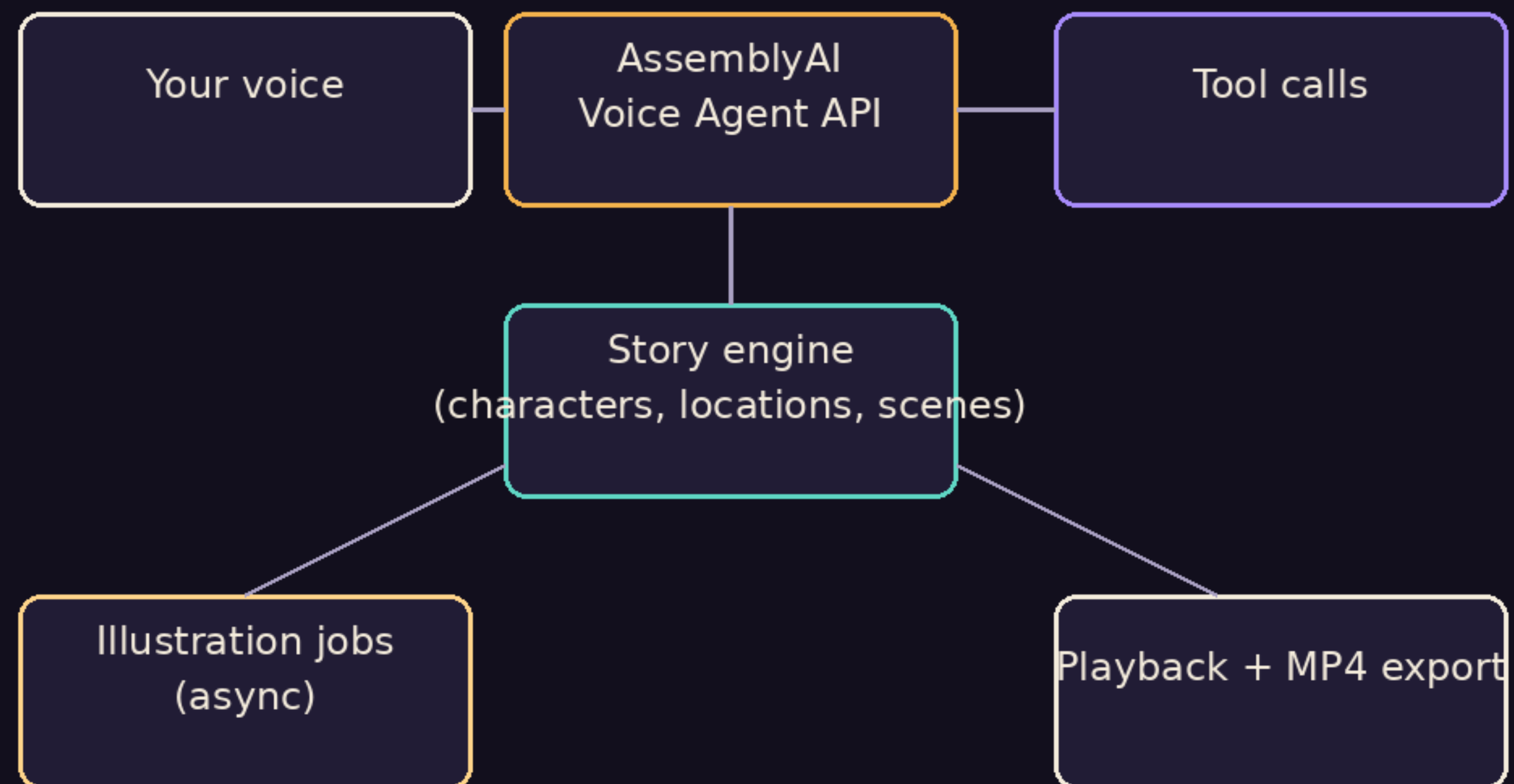
Creative ideation

Classroom visualization

- These are intended use cases, not claims of adoption or measured impact.
- No claimed users, partnerships or revenue — this is a hackathon prototype.

ARCHITECTURE

Voice in, illustrated story out



Full diagram and code layout in the README.

DEMO & LINKS

See it, run it, read it

- GitHub — github.com/NzJulien/Narraflow (MIT license, tests, CI)
- Live demo — deployed on Vercel (Sandboxes runtime); URL in the submission
- Scripted demo — </app/studio.html?demo=1>, no microphone or key needed
- 71 backend/browser tests + 37 frontend tests, all passing

Next steps

- Tune tool-calling consistency further with real speaker sessions
- Real (Fireworks FLUX) illustrations, not just the offline placeholder artist